

Sustainable approach for integrated microwave pretreatment and photocatalytic upcycling of sugarcane (*Saccharum officinarum*) leaf biomass to lactic acid

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01 INTRODUCTION

The sugarcane leaf (SCL) biomass is a promising renewable feedstock for biofuel and biochemical production. The application of photocatalysis to biorefinery is of great significance for sustainable development, however limited success has been achieved.

Research focus : To develop non-conventional and **energy-efficient** biomass valorization routes for high value lactic acid production under **sustainable process** conditions

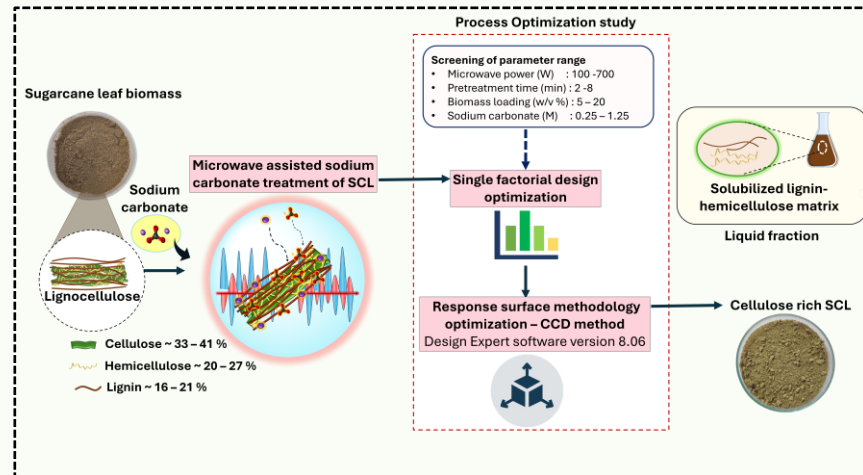
02 OBJECTIVES

- I** - To develop sustainable technologies for delignification of Sugarcane (*Saccharum officinarum*) leaf (SCL) Biomass
- II** - Photocatalytic valorization of fractionated cellulose rich SCL into high value platform chemicals – Lactic Acid (LA)
- III** - Evaluation of the Techno-economic assessment and environmental impact of the overall process for Industrial scale-up framework

03 METHODOLOGY

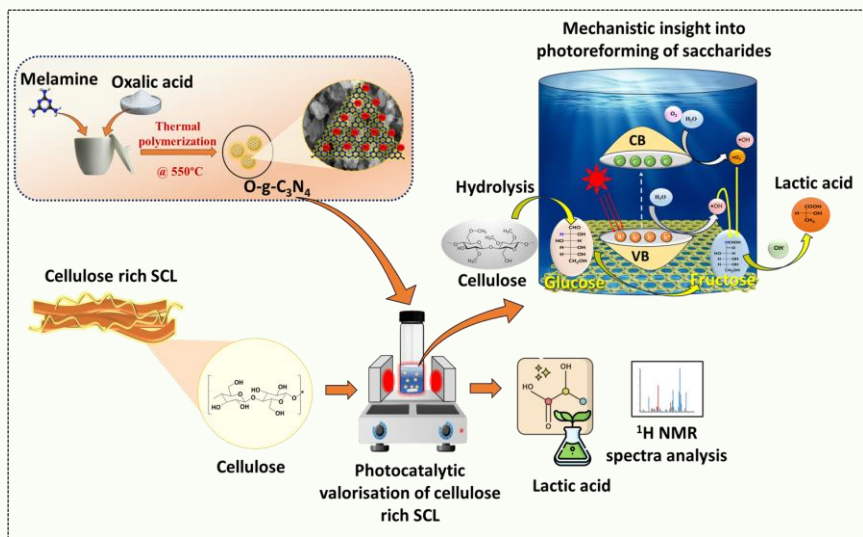
Objective I

- Stage I** - Screening of pretreatment process condition
- Stage II** - Response surface methodology (RSM) optimization
- Stage III** - Physicochemical characterization study of treated and untreated SCL biomass



Objective II

- Stage I** – Development of $g-C_3N_4-O_{50\%}$ photocatalyst and its characterization study
- Stage II** - Photocatalytic conversion of cellulose rich SCL biomass and process optimization study
- Stage III** – Separation of LA from reaction mixture



Objective III

- Stage I** - Process design and simulation (Software : SuperPro designer V10)
- Stage II** - Economic metric analysis
Capital expenditure (CAPEX)
Operational expenditure (OPEX)
- Stage III** - Sensitivity analysis
Selling cost of LA, formic acid and Lignin

04 RESULTS

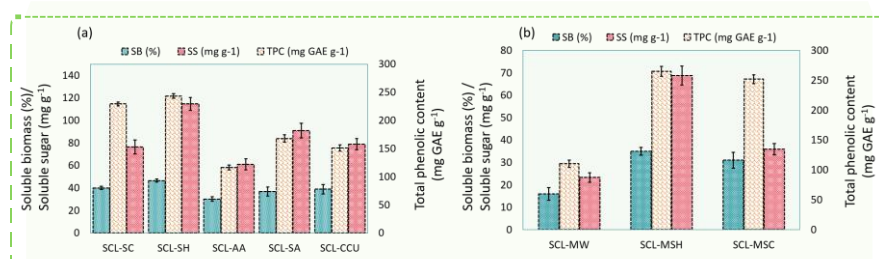


Figure 1 (a). Screening of chemical medium for pretreatment of SCL; (b) Selection of effective medium on SCL pretreatment

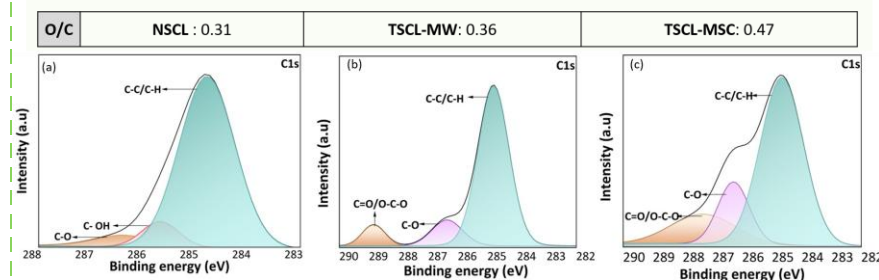


Figure 2 C1s spectra for (a) NSCL: untreated (native) SCL; (b) TSCL-MW: microwave assisted water pretreated SCL; and (c) TSCL-MSC: microwave assisted SC pretreated SCL; O/C atomic ratio on surface of SCL biomass

Table 1. Comparative analysis of saccharification yields from sugarcane leaf biomass at different optimization stages (RSY: Reducing sugar yield ; GY: Glucose yield)

Strategies	Pre-treatment condition	RSY (mg g ⁻¹)	GY (mg g ⁻¹)
Native SCL biomass	Untreated	133	56
Single factorial method	Pretreatment time : 6 min Microwave power : 500 W SC concentration : 0.75 M Biomass loading : 10 %	528	204
Central composite design method	Pretreatment time : 4 min Microwave power : 508 W SC concentration : 0.87 M Biomass loading : 7.87 %	601	231

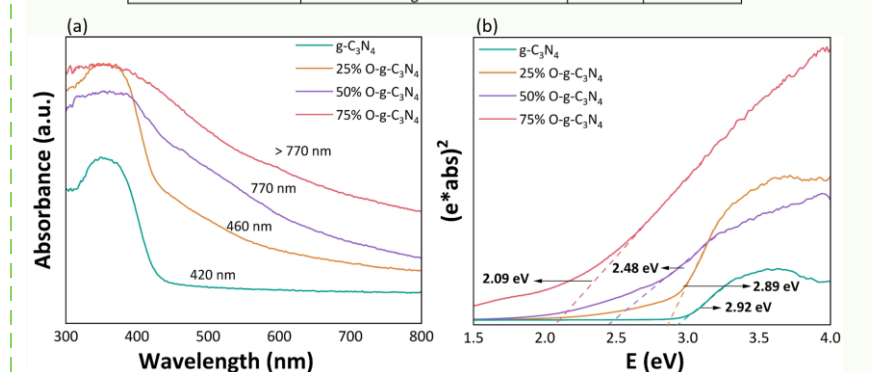


Figure 3. (a) UV-vis DRS; (b) Tauc's Plot of synthesized $g-C_3N_4-O$ and $g-C_3N_4-O_x$ % ; X : Weight percentage of oxalic acid (25 %, 50 % and 75 %)

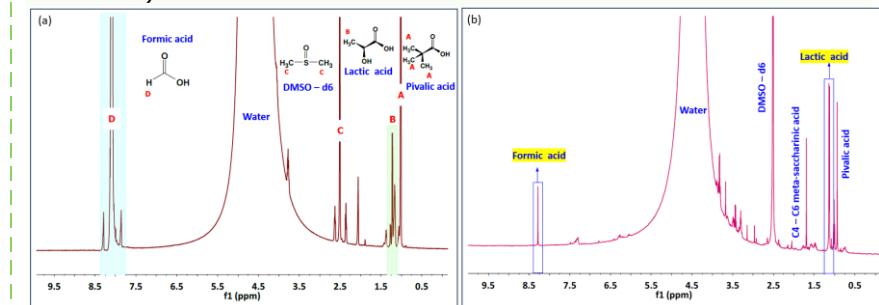


Figure 4. (a) ¹H NMR spectrum : Standard sample mixture (Lactic acid & Formic acid) ; (b) ¹H NMR spectrum : Product mixture sample

Table 2. Result overview of the organic acid yields for the effect of parameter study on reaction condition

S.No	Reaction condition	Total organic acid yield LA (mg g ⁻¹ TSCL) FA (mg g ⁻¹ TSCL)
1	Before optimization Photocatalyst : $g-C_3N_4-O_{50\%}$ Irradiation time : 8 h NaOH concentration : 1M Photocatalyst loading : 1 mg mL ⁻¹	178.64 96.50
2	After optimization : Photocatalyst : $g-C_3N_4-O_{50\%}$ Irradiation time : 16 h NaOH concentration : 2M Photocatalyst loading : 1 mg mL ⁻¹	345.39 30.45

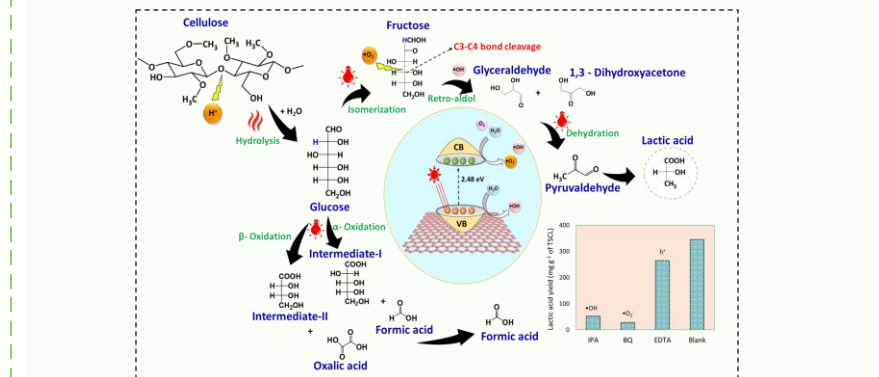


Figure 5. (a) Effect of different active oxidation species on the synthesis of LA yield; (b) Possible reaction mechanism for conversion of cellulose to organic acid synthesis

Table 3. Executive summary of the designed process for lactic acid production from SCL biomass

Total Capital Investment	27,468,000 \$
Capital Investment Charged	27,468,000 \$
Operating Cost	13,215,000 \$/yr
Main Revenue	6,373,000 \$/yr
Other Revenues	8,899,431 \$/yr
Total Revenues	15,273,000 \$/yr
Batch Size	3,808.07 kg MP
Cost Basis Annual Rate	3,983,241 kg MP/yr
Unit Production Cost	3.32 \$/kg MP
Net Unit Production Cost	3.32 \$/kg MP
Unit Production Revenue	3.83 \$/kg MP
Gross Margin	13.47 %
Return On Investment	14.43 %
Payback Time	6.93 years
IRR (After Taxes)	10.10 %
NPV (at 10.0% Interest)	165,000

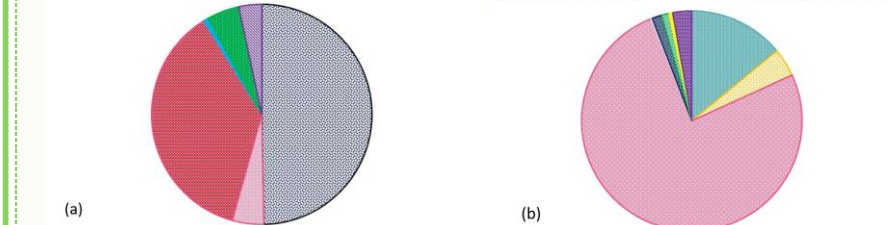


Figure 6. (a) Overall production cost for lactic acid production from SCL biomass; (b) Raw material cost for lactic acid production from SCL biomass

05 DISCUSSION

- The optimization of pre-treatment of SCL biomass under condition (Microwave power – 508W , Pretreatment time – 4 min, Biomass loading – 7.87 %, and SC concentration – 0.87 M) resulted in **RSY of 601 mg g⁻¹** and **GY of 231 mg g⁻¹**
- Similar research study stated that the interaction of the parameters such as SH concentration (2 %) and substrate loading (10 %) resulted in a high RSY of 0.529 g g⁻¹ [1]
- Photocatalytic process converted cellulose-rich SCL to **Lactic acid yield of 345.39 mg g⁻¹ TSCL and Formic acid of 30.45 mg g⁻¹ TSCL** under optimal condition (Irradiation time – 16 h ; SH concentration – 2M ; Photocatalyst $g-C_3N_4-O_{50\%}$ loading – 1 mg mL⁻¹)
- The previous study reports on the conversion of glucose to LA at 48 h under $Ba(OH)_2$ catalyst to produce a 78.3 % yield [2] and another study observed 63.65 % of LA from xylose under 10 mg of O-CNTs-2.0 photocatalyst with 1M KOH concentration [3].
- The total capital investment of \$27.47 million, with an annual revenue projection of \$ 15.27 million was primarily generated from LA production process.
- The TEA analysis shows LA separation process demands the largest portion of operational costs contributing 46.69 %
- A similar study on the techno-economic assessment of LA production from various feedstocks found that 70 % of the production cost is attributed to raw materials and facilities [4]

06 CONCLUSIONS

- The low strength alkali sodium carbonate and intensification with microwave pretreatment method remarkably promoted less degradation of cellulose in SCL
- The synergistic effect of smaller energy band (2.48 eV) of $g-C_3N_4-O_{50\%}$, prolonged irradiation time (16 h) and superoxide radicals ($\cdot O_2^-$) facilitated selective conversion of SCL biomass into LA
- The TEA analysis shows that ethyl acetate costs drives operational expenses and implementation of advanced separation processes can improve the overall economy of the process

Sugarcane leaf valorization via $g-C_3N_4-O_{50\%}$ photocatalysis, yields 345.39 mg g⁻¹ TSCL of lactic acid with demonstrated techno-economic viability.

—KEY FINDING

07 FUTURE WORK

- Hydrogen production from reaction mixture containing formic acid - To ease the separation of lactic acid
- High value phenolic chemical production from lignin derived from pretreated liquid fraction of SCL biomass

08 REFERENCES

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